

REQUIRMENT SPECIFIC / PRELIMINARIES

3.1 Existing System

Traditional radiation monitoring systems primarily rely on handheld radiation detectors or fixed monitoring stations to measure radiation levels. While these systems provide accurate radiation measurements, they do not offer real-time geospatial visualization or continuous environmental monitoring. Radiation data is often collected manually and analyzed separately, making it difficult to identify the exact location of radioactive sources during field operations. Furthermore, conventional systems lack integration between radiation sensors, GPS modules, environmental sensors, and visualization platforms, limiting their effectiveness in large-scale radiation surveillance and emergency response scenarios.

3.2 Problem Statement

The major challenge identified in existing radiation monitoring systems is the inability to visualize radiation distribution in real time while accurately identifying the source location. Conventional 2D radiation heatmaps only represent radiation intensity over an area without providing depth information or spatial context. Additionally, the absence of integrated GPS telemetry, environmental sensing, and live visualization limits decision-making during nuclear inspections and emergency situations. Therefore, an intelligent system capable of real-time radiation mapping, environmental monitoring, and future radioactive source localization is required.

3.3 Proposed Solution

The proposed solution is the development of **RADMAP**, a GPS-enabled radiation mapping prototype based on **Raspberry Pi-5**. The system integrates a **Gamma Spectrometer**, **GlobalSat BU-353N GPS Receiver**, **DHT22 environmental sensor**, and **camera module** to collect radiation counts, GPS coordinates temperature, humidity, and live video

simultaneously. The acquired data is processed through a Python-based backend to generate interactive **2D radiation heatmaps** and a **3D Digital Twin** for enhanced visualization and future radioactive source localization. A color-grading algorithm inspired by night-vision technology is implemented to represent varying radiation intensity levels on the live camera feed, providing intuitive identification of radiation hotspots. The modular architecture also supports future enhancements such as AI-based source localization, LiDAR integration, autonomous robotic navigation, and cloud-based radiation monitoring.

4.4 Requirement Specifications

Component	Specification
Raspberry Pi 5	8 GB RAM
GlobalSat BU-353N	GPS Receiver
CNSPEC MCA	Gamma Spectrometer
DHT22	Temperature & Humidity Sensor
Camera Module	Live Monitoring

Table 1: Hardware Requirements

Software	Purpose
Raspberry Pi OS	Operating System
Python 3.x	Backend Development
Streamlit	Dashboard
OpenCV	Camera Processing
Plotly	3D Visualization
Folium	2D Heatmap
Pandas	Data Processing
NumPy	Numerical Computing
GitHub	Version Control (VC)

Table 2: Software Requirements